

Elbow Muscle Force Optimization: Cost Function n=2 vs n=4

$$\text{Cost function: } u = \left(\sum_i (F_i / PCSA_i)^n \right)^{1/n}$$

Constraints: $\sum M = 0$ (moment equilibrium about elbow), $F_i \geq 0$

Initial conditions: $F_0 = (0, 0, 0)$ for both trials

Quantity	n = 2	n = 4	Difference (n4 - n2)
F1 - Brachialis (N)	46.81	62.09	+15.29
F2 - Biceps Brachii (N)	80.10	70.69	-9.41
F3 - Brachoradialis (N)	38.05	39.35	+1.30
JRF magnitude (N)	126.39	132.31	+5.91
JRF angle (deg)	-55.67	-57.06	-1.39
Cost value, u	14.21	11.00	-3.21

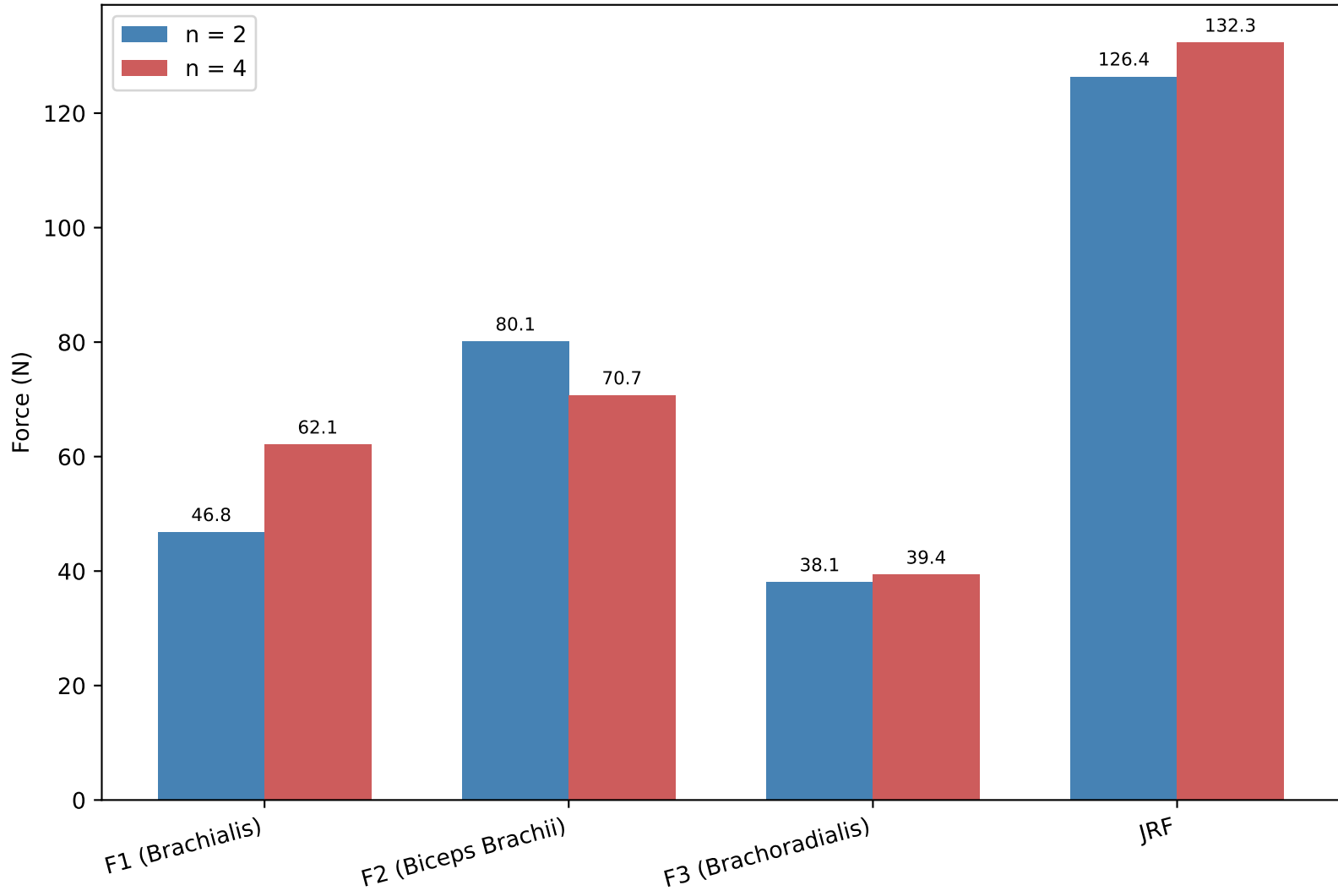
Note: u is the n-th-root-normalized (Lp-norm-style) muscle stress, so it is on the same general scale for any n, but n still changes how it weights each muscle's stress (larger n weights the most-stressed muscle more heavily), so u values are only loosely comparable across n -- the forces and JRF are the more meaningful comparison.

Interpretation

Raising the exponent n from 2 to 4 makes the cost function penalize large individual muscle stresses much more steeply. As a result, the n=4 solution redistributes load away from the muscle carrying the largest share under n=2 (Biceps Brachii, F2) and toward the muscle with the smallest moment arm (Brachialis, F1), which must work harder to help satisfy moment equilibrium.

F3 (Brachoradialis) changes only slightly since it already carries a comparatively low, moderate share of the stress. Because all three muscle forces increase in aggregate contribution to the equilibrium condition, the resulting joint reaction force is slightly larger and at a slightly steeper angle under n=4 than under n=2.

Muscle Forces (F1-F3) and Joint Reaction Force: n=2 vs n=4



Joint Reaction Force Vector: n=2 vs n=4

